Project Tilepátheia

Picking Up A Real Object By Attention Alone

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Project Flamescion turns attention alone in to a robotic retrieval. A person with severe motor impairment can still see the object they want, when the arm itself is the thing that no longer answers, and every assistive route leans on one surviving channel, a gaze that holds steady or a voice that still forms words. Our system asks for neither. Four object names are flashed across a screen, and the name which the user silently attends to draws a P300 near three tenths of a second which the other three never produce.

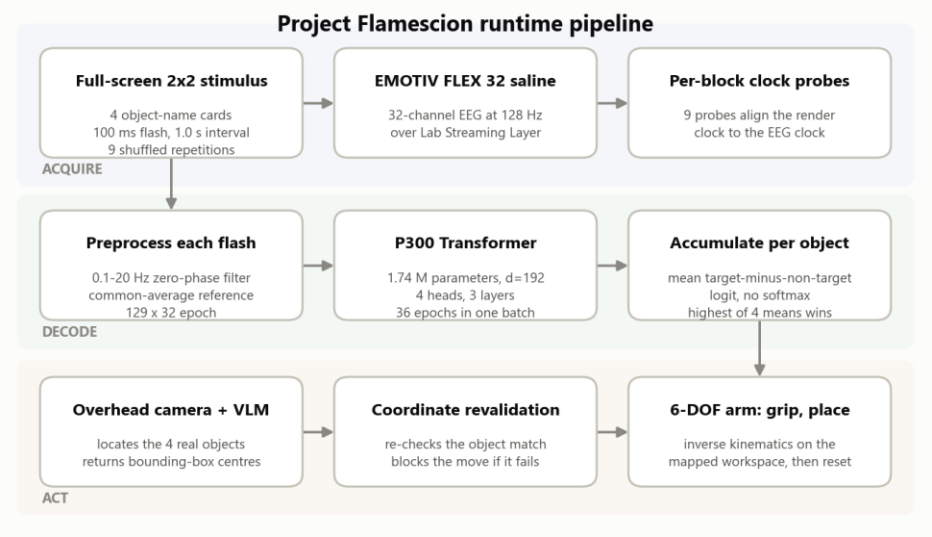


Fig 1. *One selection, flash to grip.*

The build joins four parts. A camera and vision model read the table and name up to four objects, an EMOTIV FLEX 32 saline cap streams EEG at 128 Hz over Lab Streaming Layer, a 1.74 million parameter Transformer scores each 129 by 32 epoch, and a 6 axis arm makes the pick. The network was firstly trained on public P300 corpora, then fine tuned per person over 2,205 documented runs. One flash is weak, and hence nine repetitions are pooled before anything is decided.





Fig 2a. *The four card stimulus.* **Fig 2b.** *The saline headset.*

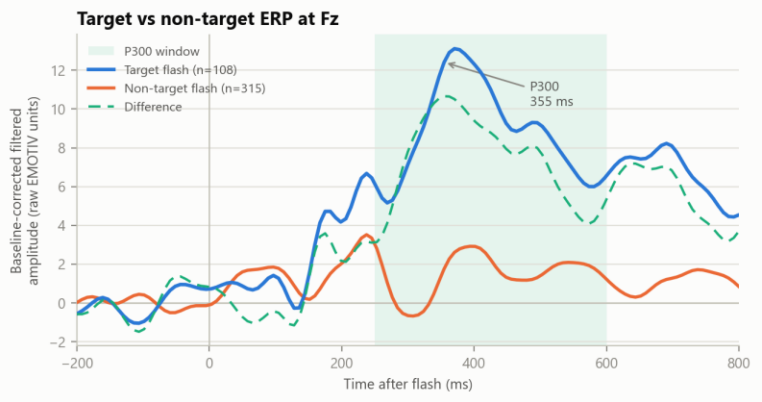


Fig 3. *Individual 3. The attended flash produce a P300.*

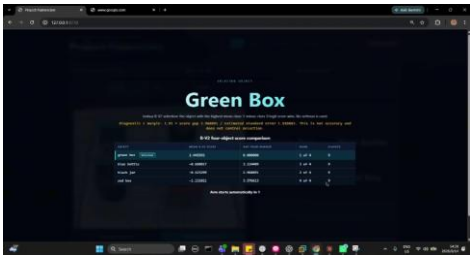
The tables report Transformer validation and diagnostic-test splits. Whole sessions are held back, since every block inside a session shares the cap placement and the state of that day, and hence a random split hands the model what it never has in use. Each checkpoint is ranked by the weaker of its two scores. Individual 3 holds 91.7% on both splits at once. Judged alone a flash is a poor classifier, when the object accuracy it supports is far higher, because we keep how strongly it responded instead of flattening it in to a vote.

Checkpoint	Validation	Held out test	Object AUC	95% CI
Alpha (deployed)	**75.0%**	**83.3%**	**0.889**	**51.6-97.9%**
Beta	66.7%	66.7%	0.819	44.7-84.4%
Gamma	62.5%	70.8%	0.806	48.9-87.4%

Table 1. *Sessions excluded from training. Chance 25%.*

Individual	Clean runs	Paired val.	Paired test	Peak val.	Peak test
Individual 1	1,140	66.7%	66.7%	83.3%	66.7%
Individual 2	1,126	66.7%	50.0%	75.0%	58.3%
Individual 3	**233**	**91.7%**	**91.7%**	**91.7%**	**91.7%**
Individual 4	37	50.0%	50.0%	66.7%	50.0%
Individual 5	1,283	75.0%	58.3%	75.0%	66.7%
Individual 6	3,085	66.7%	66.7%	91.7%	75.0%

Table 2. *Validation and diagnostic-test splits, one saved run.*



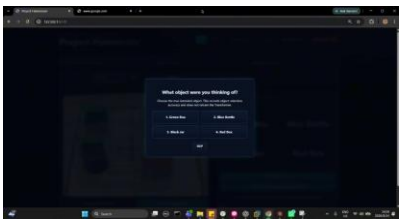


Fig 4a. *The decision screen.* **Fig 4b.** *The ground truth prompt.*



Fig 5. *The arm executing a selection.*

Beyond those splits the loop has been driven live, the arm fetching the decoded object at roughly seventy percent, a figure not tabulated at here. In use this becomes fetching and feeding for a person with ALS or a high spinal injury, and hands free work on sterile benches. Everything runs on one laptop alongst the arm, every block logged for audit, when the objects swap out without retraining since the cards carry names and not pictures.